MZUZU UNIVERSITY

SELF-RELIANCE SERVICE PERFECTION

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Department of Information & Communication Technology(ICT)

Programme Brochure

http://ict.mzuni.ac.mw

1. INTRODUCTION TO THE DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)



The Department of Information and Communication Technology (ICT) at Mzuzu University is one of the finest ICT departments in the country. We offer several modern undergraduate programmes with a host of opportunities at the undergraduate level. Our industry engaged programmes match the needs of our economy and have a reputation for producing the most employable I CT graduates in the country and region. These highly skilled graduates are in huge demand and contribute significantly to the development of the country. As technology plays a greater role in our society the growth in the demand for these graduates continues year after year.

Our programmes teach students how to think critically and apply logical reasoning to solve complex problems. Students learn key transferable skills that make them both competitive and adaptable when they enter the workforce. The increasing demand for our graduates has led to above average and faster wage growth in this field relative to others. There are many exciting industry and research opportunities for graduates in roles such as Software Development, Cybersecurity, System Administration, Big Data, Data Science, Internet of Things (IoT), Research, IT Management and many more.

The field of ICT is always growing and changing. Our cities, homes, cars and accessories will become smarter as they connect to the Internet and each other. New and improved computing technology will continue to create new opportunities and challenges. Our graduates have the skills to meet these challenges and opportunities. Staff from the Department of ICT work closely with the industry to create programme content that's current and relevant. As a result, students gain relevant skills and knowledge that will benefit them for years to come.

Today, there are two main branches of Computer Science and we offer programmes for both:

- Data Science programmes teach you to analyse, understand, manage, interpret and present data.
- IT programmes teach you how to work in and/or manage an IT department in a company and how to develop, set up or maintain existing systems and services.

Scholarships for Our Programmes

Several scholarships are available post first year. These include the Higher Education Loans Board.

2. DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Why should you study at Mzuni?

Highly Respected Qualification: Mzuzu University is internationally recognised for its excellence in research, learning and engagement with industry.

Small lab groups: We operate small lab groups to ensure no one gets lost in the crowd and everyone can get the help they need to progress.

Focus is on labs/practicals: 50% to 80% of time spent with the lecturer on each module is in labs. The best way to learn is by doing.

Fundamental to Advanced: In the first year, you'll learn the basics of several ICT topics. Each semester, you'll build on that knowledge to an advanced level by the final year.

Quality: The ICT lecturers are highly-qualified experts and researchers in many areas including:

- Programming (Java, Python, C, PHP, and JavaScript - the languages that allow you to create applications to do things)
- Data Analytics (making sense of the information in our world)
- Internet of Things (creating smart everyday items)
- Information Security (keeping your private information safe)
- Artificial Intelligence
- Machine Learning (making computers/machines think like us)
- Cybersecurity

Progression: Whether you start at Diploma Level, you'll have the opportunity to progress to the fourth year (BSc) within the department.

Work placement: This is organised through the department for 4th-year students with local and national companies. You'll get hands-on experience in the industry, your first professional job reference, perhaps an exciting Final Year Project with a company, or maybe even a future job opportunity. Some of the companies recently involved in providing work placement opportunities are:

Job Opportunities: Recent graduates are working in various companies while others are on internship. The other graduates have established companies of their own.

Flexibility: It's difficult to be certain of the programme you want to complete before you start it. There's a large amount of flexibility for students to change programmes after the first year because the first-year curriculum is common for all degrees in the Department of ICT.

UNDERGRADUATE DEGREES

DIPLOMA IN ICT
(Weekend Mode)

Overview

This is an undergraduate degree programme with a variety of practical and theoretical courses. As a requirement towards partial fulfilment for the award of the degree, the students will complete a research project to demonstrate the application of knowledge and skills acquired throughout the tuition. Furthermore, all students will complete a mini-project during the second year as a requirement. It is 2-year face-to-face multiple entries and multiple exit programme. The students enter the programme in Year 1 (Level 1).

Curriculum

One of the benefits of this diploma is that it has a range of modules that are modern, relevant and interesting. These modules cover current and future trends in the industry. The main focus of the diploma is on programming, so you will learn languages such as Python, Java, C, JavaScript and PHP. You will also learn about Databases, NoSQL, Networks and Server Administration, and Systems Analysis and Design. You will complete projects on your own and in groups throughout the study, which will prepare you for working in the industry.

Career Opportunities

You will have career opportunities in Malawi and abroad in various areas. Career options for graduates include; Software Developers, Systems Analysts, Database Managers, Network Technicians, Hardware Technicians and other diverse range of roles.

Further Studies

There is an opportunity to upgrade to a degree level in the Department of ICT through BSc ICT (Upgrading). Check the next section.

Contact Information

Mr Emmanuel Ngalande
Department of ICT
T: +265 (99) 556 2153
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Admission

For admission to the programme, standard applicants must

- Possess a Malawi School Certificate of Education (MSCE) or its equivalent obtained in not more than two sittings with at least six (6) credits.
- meet the minimum entry requirements

Minimum Entry Requirements MSCE with at least 6 credits	
Mandatory Subjects	Subjects include
4	English, Mathematics/Additional Mathematics, Biology, Physical Science/General Science/(Physics and Chemistry) and any two science-related subjects.

Mode of Delivery

The mode of delivery is face to face but on a weekend basis.* The programme is offered during weekends as follows:
Friday to Sunday every other week (fortnightly)
Friday 3:00 pm – 9:00 pm = 6 hours
Saturday 7:00 am – 12:00 pm = 5 hours
1:00 pm – 6:00 pm = 5 hours
Sunday 7:00 am – 12:00 pm = 5 hours

More information:
http://ict.mzuni.ac.mw/?page_id=2347

UNDERGRADUATE DEGREES

BSC IN ICT (UPGRADING)
(Weekend Mode)

Overview

This is an undergraduate degree programme with a variety of practical and theoretical courses. As a requirement towards partial fulfilment for the award of the degree, the students will complete a research project to demonstrate the application of knowledge and skills acquired throughout the tuition. In addition to the research project, the students will undergo an assessed industrial attachment. Furthermore, all students will complete a mini-project during the third and fourth year as a requirement. It is three-year face-to-face multiple entries and multiple exit programme. The students enter the programme in Year 2 (Level 2) or Year 3 (Level 3). It blends theory and practice into a learning experience that develops skills applicable to complex real-world problems in the field of ICT.

Curriculum

The programme covers a wide range of modules that will be useful in any type of development roles such as programming, networking, databases, web development, app development, server administration, modelling and simulation, Information Systems management, web applications security, and Artificial Intelligence. This produces a graduate who can take a project from the initial concept/design stage right through to delivering the final product. You will complete projects on your own and in groups throughout the study, which will prepare you for working in the industry.

Career Opportunities

Graduates of this programme will be specialists in Information systems for companies, Network systems design and implementation, Information systems security, Software Application Design and development, Distributed systems design and development, Digital security and artificial intelligence.

Further Studies

Graduates from this programme qualify to compete and pursue postgraduate opportunities in any prestigious university around the globe.

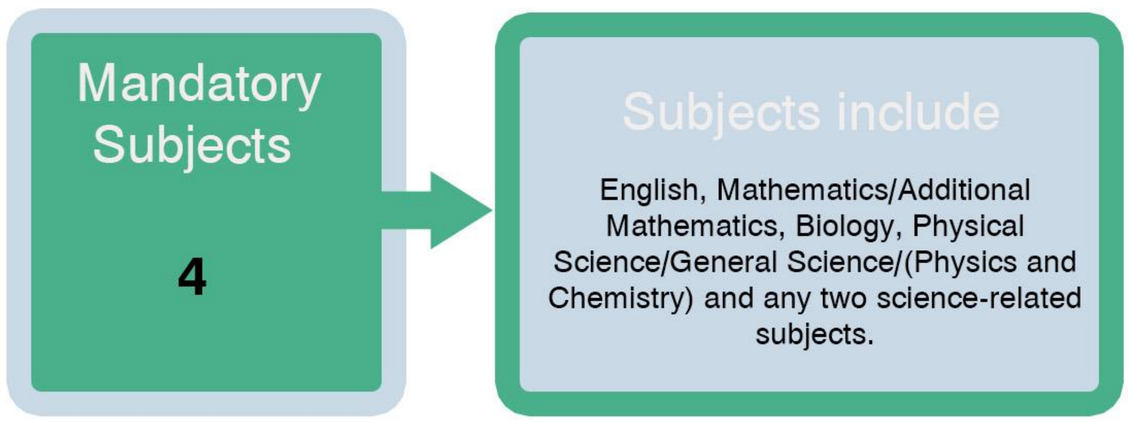
Contact Information

Mr Emmanuel Ngalande
Department of ICT
T: +265 (99) 556 2153
E: ngalande.e@mzuni.ac.mw

Admission

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- Possess a Malawi School Certificate of Education (MSCE) or its equivalent obtained in not more than two sittings with at least six (6) credits.
- Meet the minimum entry requirements



Mode of Delivery

The mode of delivery is face to face but on a weekend basis.* The programme is offered during weekends as follows:
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More information:
http://ict.mzuni.ac.mw/?page_id=2341

UNDERGRADUATE DEGREES

BSC IN ICT
(Generic)

Overview

This programme offers students a rich learning experience that blends theoretical, practical and industrial skills. Its structure and delivery methods equip students with analytical and problem-solving skills in the fields of Networking, Programming, Databases, Software Development and Algorithms. This is an undergraduate degree programme with a variety of practical and theoretical courses. As a requirement towards partial fulfilment for the award of the degree, the students will complete a research project to demonstrate the application of knowledge and skills acquired throughout the tuition. In addition to the research project, the students will undergo an assessed industrial attachment. The attachment exposes the students to the real work environment. Furthermore, all students will complete a mini-project during the third and fourth year as a requirement. It is a four-year face-to-face multiple entries and multiple exit programme. The generic students enter the programme in Year 1 (Level 1).

Curriculum

The programme covers a wide range of modules that will be useful in any type of development roles such as programming, networking, databases, web development, app development, server administration, modelling and simulation, Information Systems management, web applications security, and Artificial Intelligence. This produces a graduate who can take a project from the initial concept/design stage right through to delivering the final product. You will complete projects on your own and in groups throughout the study, which will prepare you for working in the industry.

Career Opportunities

Students obtain advanced skills in Multimedia Technologies, Computer Security and Mobile Applications Development. Career options to graduates include; Software Developers, Systems Analysts, Database Managers, IT Managers, Network Engineers, Data Analysts, Researchers in in Computing and several other computing careers.

Further Studies

Graduates from this programme qualify to compete and pursue postgraduate opportunities in any prestigious university around the globe.

Contact Information

Mr Reuben Moyo
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Admission

For admission to the programme, standard applicants must

- Possess a Malawi School Certificate of Education (MSCE) or its equivalent obtained in not more than two sittings with at least six (6) credits.
- Meet the minimum entry requirements.

Minimum Entry Requirements
MSCE with at least 6 credits

Mandatory Subjects	Subjects include
4	English, Mathematics/Additional Mathematics, Biology, Physical Science/General Science/(Physics and Chemistry) and any two science-related subjects.

Mode of Delivery

Face to face; Monday to Friday.

More information:

http://ict.mzuni.ac.mw/?page_id=2324

UNDERGRADUATE DEGREES

BSC IN DATA SCIENCE (Generic)

Overview

This programme offers students a rich learning experience that blends theoretical, practical and industrial skills. Its structure and delivery methods equip students with analytical and problem-solving skills in the fields of Data Science. This is an undergraduate degree programme with a variety of practical and theoretical courses. As a requirement towards partial fulfilment for the award of the degree, the students will complete a research project to demonstrate the application of knowledge and skills acquired throughout the tuition. In addition to the research project, the students will undergo an assessed industrial attachment. The attachment exposes the students to the real work environment. Furthermore, all students will complete a mini-project during the third and fourth year as a requirement. It is a four-year face-to-face multiple entries and multiple exit programme. The generic students enter the programme in Year 1 (Level 1).

Curriculum

The programme covers a wide range of modules that will be useful in any type of big data roles. The Bachelor of Science (Data Science) programme aims at providing students with scientific knowledge, analytical and interpretational skills, values and attitudes required for them to competently apply data science concepts in solving problems. You will complete projects on your own and in groups throughout the study, which will prepare you for working in the industry. Intelligence Analyst, Data Architect, information analysts.

Career Opportunities

When you finish the course, you'll be equipped for a career in the data analytics industry. You could work for multinational companies, small and medium enterprises, and innovation centres. You can do the following jobs: Data Scientist, Process Analyst, Database Administrator, Data Solutions Analyst, Business Analyst, Statistical Analyst, Data Mining Engineer, Web & Social Media Analyst, CRM Analyst, Business Intelligence Analyst, Data Architect, information analysts.

Further Studies

Graduates from this programme qualify to compete and pursue postgraduate opportunities in any prestigious university around the globe.

Contact Information

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E: namacha.e@mzuni.ac.mw

Admission

For admission to the programme, standard applicants must

- Possess a Malawi School Certificate of Education (MSCE) or its equivalent obtained in not more than two sittings with at least six (6) credits.
- Meet the minimum entry requirements.

Minimum Entry Requirements MSCE with at least 6 credits	
Mandatory Subjects	Subjects include
4	English, Mathematics/Additional Mathematics, Biology, Physical Science/General Science/(Physics and Chemistry) and any two science-related subjects.

Mode of Delivery

The mode of delivery is face to face but on a weekend basis.* The programme is offered during weekends as follows:
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More information:
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PROFESSIONAL CERTIFICATES

Professional Training Opportunities

We also provide a variety of demand-driven certification programmes. We are an Accredited Training Centre for EC-Council, CISCO, and International Computer Driving License (ICDL)

EC-COUNCIL

- Certified Ethical Hacker (CEH)
- Certified Network Defender(CND)*

CISCO

- Cisco Certified Network Associate (CCNA)

International Computer Driving License (ICDL)

The Department of ICT is an Accredited Training Center for ICDL courses from ICDL Workforce Modules to ICDL Professional Modules.
Evening Classes are offered to support even those who go to work.

Contact Information

Ms Florence Nkosi
Department of ICT
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E: nkosi.f@mzuni.ac.mw

Please visit http://ict.mzuni.ac.mw/?page_id=2365 for more details.

In the Department of ICT, there are many paths and possibilities to pursue a programme that suits you, and to excel in it with the expert help and support from staff!